

Verified Generators and Relations for Qupit Clifford Operators

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Bian, Li, Ross, van de Wetering, and Zhao give sixteen rewrite rules that are sound and complete for multi-qudit Clifford circuits over H, S, CZ in every odd prime dimension p . We report a machine-checked proof of the result in safe, axiom-free Agda, against an independently constructed semantics. Rather than follow the paper, which applies one rule set to the whole Clifford group “up to Paulis”, we factor the projective Clifford group as $\overline{\mathcal{P}}_n \rtimes \text{Sp}(2n, \mathbb{Z}_p)$, present the factors separately, and glue the presentations with a generic, verified semidirect-product theorem. The symplectic factor carries the mathematics: the paper’s normal form, refined into a doubly inductive datatype of *normal boxes* and *rows*, is a tower of verified Reidemeister–Schreier coset tables, proved unique by induction on the width, and the paper’s 66 box relations are derived from eighteen natural ones. Verified isomorphisms of presentations reach the paper’s own rules, and a central extension, verified against a cocycle, restores the scalars. Everything is parametric in p and typechecks with no postulates.

CCS Concepts: • **Theory of computation** → **Type theory; Logic and verification; Equational logic and rewriting; Quantum computation theory.**

Additional Key Words and Phrases: Clifford group, qudits, generators and relations, completeness, normal forms, symplectic group, semidirect product, Agda, formal verification

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1 Introduction

A finite list of circuit equations is *sound* when each holds in the intended semantics and *complete* when every semantic equality between circuits is derivable from it. Completeness is proved by a *normal form*, and the case analysis of every generator against every normal form is enormous: Selinger’s n -qubit Clifford presentation [Selinger 2015] runs a substantial rewrite system through it, its qutrit successor [Li et al. 2025] needs over fifty pages, and its generalisation to all odd primes by Bian, Li, Ross, van de Wetering, and Zhao [Bian et al. 2026a] — sixteen rules for multi-qudit Clifford circuits in every odd prime dimension p , henceforth “the qupit paper” — rests on roughly a hundred pages of derivations [Bian et al. 2026b].

We report a machine-checked proof of that theorem in safe, axiom-free Agda [Norell 2007; The Agda Community 2026], against a semantics constructed independently of the syntax, on the library of [Anonymous 2026a,b] (introduced in Appendix C). The proof does *not* follow the paper’s, which works *up to Paulis* with one rule set for the whole group (§3); we instead *factor the group, present the factors, compose the presentations*: $\overline{C}_n = \overline{\mathcal{P}}_n \rtimes \text{Sp}(2n, \mathbb{Z}_p)$ and C_n is a central extension of it by the scalars; the symplectic group gets a coset-table normal form, the Pauli group a compositional presentation, the scalars a cyclic one, generic verified theorems for semidirect products and central extensions glue them, and plumbing reduces the result to the paper’s H, S, CZ and rules.

2 Qupit Clifford Circuits and the Main Theorems

Fix an odd prime p and $\omega = e^{2\pi i/p}$. On the basis $|0\rangle, \dots, |p-1\rangle$ indexed by $\mathbb{Z}_p$, $X|j\rangle = |j+1\rangle$ and $Z|j\rangle = \omega^j|j\rangle$ generate, with the phases ω^t , the Pauli group $\mathcal{P}_n$, and $H: |j\rangle \mapsto \frac{\delta_p}{\sqrt{p}} \sum_k \omega^{jk}|k\rangle$,

$S: |j\rangle \mapsto \omega^{j(j-1)/2}|j\rangle$, $CZ: |j\rangle|l\rangle \mapsto \omega^{jl}|j\rangle|l\rangle$ generate, with the phases ω^t , the Clifford group C_n ($\delta_p = i^{(p-1)/2}$ makes the determinant of H equal to 1; the determinants of S and CZ are 1 too) [Bian et al. 2026a]. Modulo scalars, $\overline{C}_n = C_n/\langle\omega\rangle$ and $\overline{\mathcal{P}}_n \cong \mathbb{Z}_p^{2n}$, $C_n/\mathcal{P}_n \cong \overline{C}_n/\overline{\mathcal{P}}_n \cong \text{Sp}(2n, \mathbb{Z}_p)$, and $\overline{C}_n \cong \overline{\mathcal{P}}_n \rtimes \text{Sp}(2n, \mathbb{Z}_p)$. These “glue facts” are the two not formalised: our semantic groups are defined as the right-hand sides. In [Bian et al. 2026a] the global phases also include -1 , so that the swap and the multiplier gates belong to C_n ; we work mainly in the projective quotient, where -1 makes no difference except that the relations come out a little simpler.

Circuit $n = \text{Word}(\text{Gen } n)$ is the free monoid on wire-indexed generators over the gates H , S (unary) and CZ (binary), and $_ \uparrow$ shifts a circuit up one wire. The other usual gates are *words* ($Z = HSHHS^{-1}$, $X = HSHS^{-1}H$, the swap $Ex = (CZ \cdot H \downarrow \cdot H \uparrow)^3$), as is the *multiplier* $M_x|j\rangle = |xj\rangle$, $x \in \mathbb{Z}_p^*$, spelled by the paper over S with a Pauli prefix (its Lemma 2.14) and by us mostly as $R^x H R^{x^{-1}} H R^x H$ with $R = S \cdot Z^{1/2}$. Our primary rule set — Figure 1 modulo scalars, multiplier over R , $(SH)^3 = \varepsilon$ kept as an axiom — is one inductive family $_ \text{CRel}_{===}$ with sixteen constructors, six one-qubit (orders, the multiplier calculus M-power and semi-MR, comm-HSHHS), seven two-qubit (orders of CZ and Ex , commutations, the swap rules, blake-c12) and three three-qubit (Yang–Baxter, cz-slide, semi-CX $\uparrow$ -CZ $\downarrow$), all drawn in Figure 1, plus the three *structural* rules every circuit theory gets free (congruence under $\uparrow$; gates on disjoint wires commute); $XZ = ZX$ modulo scalars, Ex commuting with CZ and both rules pushing a Pauli through CZ are derived. Figure 1 has fifteen non-scalar rules; under the R -spelling $(SH)^3 = \varepsilon$ must stay an axiom, and §5 verifies the paper’s fifteen in its S -spelling, where it is a theorem.

THEOREM 2.1 (CLIFFORD-PRESENTATION). *For every width n , the sixteen rules present the projective Clifford group:*

Theorem-1 : $\forall (n : \mathbb{N}) \rightarrow (n \text{ CRel}_{===}) \text{ IsPresentationOf } (\text{Pauli} \times \text{Sp } n)$

Theorem-1 = *PapPres.presentation*

$\text{Pauli} \times \text{Sp } n$ is the semidirect product of $\mathbb{Z}_p^{2n}$ by $\text{Sp}(2n, \mathbb{Z}_p)$ (§5); $_ \text{IsPresentationOf}$ (Appendix C) is a group isomorphism $\text{Cir } n / \approx \rightarrow \overline{\mathcal{P}}_n \rtimes \text{Sp}(2n, \mathbb{Z}_p) \cong C_n / \langle\omega\rangle$: *soundness*, each rule holds, and *completeness*, $[[_]]$ is injective. Normal forms for symplectic operators (Clifford operators modulo Paulis) are *data*: four kinds of atomic box, the paper’s (its Figure 6), $A = \{(a, b) \in \mathbb{Z}_p^2 \mid (a, b) \neq (0, 0)\}$, $B = D = \mathbb{Z}_p^2$, $E = \mathbb{Z}_p$, spelled as circuits in Figure 2, then rows of boxes and a staircase of rows (§4).

THEOREM 2.2 (UNIQUE-NF). *Normal forms with equal symplectic denotations are equal:*

Theorem-2 : $\forall n \{u \vee : \text{NF } n\} \rightarrow [[[u]]] \approx^s [[[v]]] \rightarrow u \equiv v$

Theorem-2 = *U.[[]]-injective*

3 Two Routes to Completeness

The routes share the normal boxes, the symplectic normal form and essentially the same eighteen symplectic relations, and differ in the *organisation* of the argument. The paper’s normal form is a *symplectic normal form* followed by a *Pauli normal form*, unique by Proposition 3.8 (“analogously to the qutrit case”); Section 4.1 pushes the gates in one by one — a gate through a normal box yields an updated box and residual *dirty* gates — collecting the cases into 66 parametrised *box relations* (Theorem 4.4 reduces them to 18, a step already in Agda); relations are applied *up to Paulis*; and Proposition 4.9 boosts by correcting each rule $C_l = C_r$ to $C_l = C_r ; P$ for the unique Pauli P making it projectively sound and replaying symplectic rewrites with the corrected rules, pushing each new Pauli to the end, where it becomes “a potentially different Pauli”. *Why not formalise that*: the procedure is not a function — formalised, it is a coset table, as we write, which would also have to say

what happens to the Pauli — and the objects rewritten are circuits modulo Paulis with no syntactic representation of the quotient, so the honest counterpart of Proposition 4.9 is a normaliser on the semidirect product whose well-definedness (congruent inputs give equal cosets and congruent residuals) is exactly what “potentially different Pauli” glosses over; we did not see how to verify it at reasonable cost. *Our route:* since $\overline{C}_n = \overline{P}_n \rtimes \text{Sp}(2n, \mathbb{Z}_p)$ and C_n is a central extension of it, we work in a syntax where Paulis are genuinely trivial — symplectic circuits have only H, S, CZ and denote $\text{Sp}(2n, \mathbb{Z}_p)$ itself — and reintroduce Paulis and scalars by construction (Figure 3 in the appendix): $\text{Sp}(2n, \mathbb{Z}_p)$ is presented by a coset tower whose cosets are the normal boxes as a doubly inductive type, whose table is “push a gate through a box” as a structurally recursive function with a verified section, and whose well-definedness is proved *semantically*, by uniqueness one level down; $\overline{P}_n$ is glued on by the semidirect theorem, whose one new obligation, that the action respects both factors’ rules, follows from soundness and completeness of the factors; plumbing reduces alphabet and relations to the paper’s; and the extension theorem restores the scalars. The Pauli bookkeeping of Proposition 4.9 is thus done once, by a theorem proved for arbitrary groups before this project began.

4 The Symplectic Factor

$\text{Sp}(2n, \mathbb{Z}_p)$ is modelled as the group of $\mathbb{Z}_p$ -linear, form-preserving bijections of the phase space Pauli $n = \text{Vec}(\mathbb{Z}_p \times \mathbb{Z}_p)$ n (the exponents of $Z^a X^b$ up to phase, per wire), with extensional equality, so that the monoid laws hold by refl; the generators act as the paper’s conjugation tables (Lemmas 2.22–2.23), and soundness of every axiom is one clause per axiom. The paper’s symplectic normal form is a Z -normal then an X -normal layer on n wires, then a normal form on $n - 1$; we refine it into a doubly inductive datatype (Appendix B). An M -row is a column of D boxes ending in an E box (the X -normal layer), an L -row a column of B boxes ending in an A box (the Z -normal layer), and an ML box at width $m \geq 2$ is *either* a pair of rows *or* a bottom D box followed by an ML box one wire up: a spine of D boxes capped by two rows. ML makes explicit the position of the paper’s A box and, crucially, makes the coset type at width m a *sum* of the coset type at width $m - 1$ and a flat case, the shape the table and the uniqueness proof recurse on.

The table. The coset table is the paper’s “push a gate through a normal box” as a function $\text{ract} : ML(1+n) \rightarrow \text{Gen}(1+n) \rightarrow \text{Circuit } n \times ML(1+n)$ (5 600 lines), with soundness law ract-sound , a chain of box relations and structural rules per clause. The difference from S_n is the hypothesis that the table respects the axioms, which does *not* close by evaluation: its *coset half* is semantic — apply $[[_]]$ to ract-sound on both sides of an axiom, note that residuals live one wire up and cannot move the bottom wire, and conclude by head-injectivity of the coset section (below) — and its *residual half* follows by cancellation, given that $\uparrow$ is injective on congruence classes one width down, a consequence of completeness there. So the tower is a well-founded induction on the width.

Uniqueness. Theorem 2.2 is a structural induction on the staircase from two lemmas: *tail surjectivity*, free since a symplectic transformation carries its inverse, and *head injectivity* — ML boxes agreeing on the bottom-wire output at every input are equal — the crux, an induction on ML (LMHeadInj, 1 300 lines), the D box of a spine being read off by probing with Z and X on the bottom wire.

Relation reduction. The relations ract-sound uses are the paper’s box relations, nine families of case tables quantified over the box labels (42 parametric relations against the paper’s 66, differing only in grouping), derived (BR, 18 000 lines with its supporting calculi) from the *full* symplectic rule set of seventeen axioms: Selinger’s c10–c15 [Selinger 2015] verbatim on two and three wires, and on one wire the orders of S, H and $SH, H^2S = SH^2$ and four multiplier *schemes* over units such as $M_x M_y = M_{xy}$. The *simplified* set derives all of these from relations about the single multiplier

M_g of the primitive root (M-power $M_g^k = M_{g^k}$, semi-MS $M_g S = S^{g^2} M_g$, M_g past CZ on either wire, $H^2 = M_{-1}$; Appendix B), with c10–c15 and the orders of S and CZ: fifteen rules, eighteen with the structural ones; the identity on words is an isomorphism both ways (Simplified.Iso, 900 lines; the paper’s Theorem 4.4). The odd prime shows only on one wire; the two- and three-wire rewriting is Selinger’s. Completeness follows by by-normalization, surjectivity is “clear the bottom wire, recurse” (Theorem-LM, the paper’s Lemma 3.7); and Theorem-Sp transports the presentation to the simplified rules along the identity isomorphism.

5 Composing: Pauli, the Semidirect Product, and Scalars

$\overline{\mathcal{P}}_n \cong (\mathbb{Z}_p \times \mathbb{Z}_p)^n$ is presented over X, Z on each wire by $X^p = Z^p = \varepsilon$ and $ZX = XZ$, with the phase space that $\text{Sp}(2n, \mathbb{Z}_p)$ acts on as its semantic group and normal form $X^a Z^b$ per wire. The semidirect product is the wreath example with $\text{Sp}(2n, \mathbb{Z}_p)$ for S_n : both factors’ relations plus $h n h^{-1} = \text{conj } h n$, conj being the paper’s Lemmas 2.22–2.23 as words. Its two conditions are proved by semantics: the conjugates of congruent words have equal denotations by symplectic soundness, hence are congruent by Pauli *completeness* (respects- Δ ; the paper’s “Pauli complete” rules, Definition 4.7). Every premise being a theorem, the semidirect presentation of Pauli $\times$ Sp is unconditional.

Plumbing. The semidirect presentation has X, Z, H, S, CZ and 3+15+14 relations (Pauli, symplectic, conjugation); Simplified-V1 has H, S, CZ and nineteen; Paper-V0 and Paper-V1 sixteen and fifteen (generator reduction: X, Z become words; relation reduction: Pauli and conjugation rules become theorems). *Semidirect to V1* is an isomorphism on *different* alphabets, f sending S to R and h sending S to $Z^{-1/2} \bullet S$, because V1 spells $M_x = R^x H R^{x^{-1}} H R^x H$ exactly; V1’s nineteen rules add $(SH)^3 = \varepsilon$, $H^2 S H^2 S = S H^2 S H^2$ and the two X -through-CZ rules to the simplified symplectic ones, and f -well-defined derives the three Pauli and fourteen conjugation rules (6 000 lines). *V1 to Paper-V0* is the identity on words: ten axioms are shared, V0’s Lemmas (4 750 lines) derives c10–c15 from its own axioms. *V0 to the paper’s fifteen* connects the routes: Figure 1 spells $M_x = \text{SHS}' x^{-1} x$ with $\text{SHS}' a b = Z^{(b-1)/2} X^{(1-a)/2} S^b H S^a H S^b H$ and semi-MR as $M_g S^{g^2} Z^{(g^2-g)/2} = S M_g$; Paper-V1 restates the four multiplier rules over SHS' and has fifteen rules, since under the S -spelling M_1 is $(SH)^3$ and M-power at $k = 0$ gives $M_1 \approx \varepsilon$, whereas $(RH)^3$ is a different word; the bridge $R^a H R^b H R^a H \approx \text{SHS}' a b$ (Shared.PauliBase) carries the multiplier rules across. Hence Theorem 2.1.

Scalars. The projective presentation is extended — add ω and $\omega^p = \varepsilon$, let ω commute with every gate, *correct* each rule by the scalar it picks up — by extension-presentation with a correction that is ε except on order-SH: with H normalised by δ_p exactly one rule fails on the nose, $(SH)^3 = \varepsilon$ picking up $\omega^{(p^2-1)/8}$ (residue $-1/8$), so it becomes $(SH)^3 = \omega^{-1/8}$ (the multiplier rules hold exactly because R has no Pauli part). The semantic extension is the Weyl cocycle $c((P, S), (Q, _)) = \frac{1}{2} \text{sform}(P, \text{ap } S \ Q)$ on the natural semidirect product, whose cocycle law is symplectic invariance of sform ; the presented group’s cocycle is pulled back from it, so the two agree by construction. On no hypothesis, presentation-exact gives, for every n , $(n \text{ Exact, } _ == _)$ IsPresentationOf (Presented-group n (exact-generator-data n)); it needs no normal forms, only that each rule’s Weyl phase on the left equals the correction plus that on the right, nineteen $\mathbb{Z}_p$ equations. The extension is provably non-split.

6 Discussion, Related Work and Conclusion

Everything above is checked by Agda 2.8 with standard library 2.4 under `-safe`, no postulates, no holes. Not formalised are the *glue facts* — that the constructed groups are the groups of unitaries generated by H, S, CZ modulo (or not) phases, the paper’s Proposition 2.25 and Theorem 2.26 after Weil; no matrix appears in the development — the isomorphism of the two *total* central extensions, and the existence of primitive roots. The development is 106 388 lines in 329 files

(Table 1): framework 15 000, qupit-specific 60 000, of which the symplectic factor is 35 000 and the composition step 23 000. *Lessons*: prove well-definedness by semantics; transport, do not re-prove; make the semantics compute. A language model helped write the Agda code and drafted this paper from the authors' outline.

Related work. Presentations of gate groups with hand-checked normal forms run from Selinger's n -qubit Clifford group [Selinger 2015] through [Amy et al. 2018, 2024; Bian 2023; Bian and Selinger 2021, 2023a,b; Greylyn 2014; Li et al. 2021; Makary et al. 2021] to the qutrit [Li et al. 2025] and qupit [Bian et al. 2026a] rules; the PROP-based programme [Blake 2026a,b,c; Clément 2026; Clément et al. 2024a,b, 2023], after Lafont [Lafont 2003], and the complete ZX calculi [Backens 2014; Booth and Carette 2022; Jeandel et al. 2018; Poór et al. 2023] are pen-and-paper too. Mechanised completeness is rare (Π as a free symmetric rig groupoid [Choudhury et al. 2022] and its square-root extension [Carette et al. 2024] are coherence arguments); verified compilers and logics [Chareton et al. 2021; Hietala et al. 2021; Lehmann et al. 2025; Li et al. 2020, 2022; Paykin et al. 2017; Peduri et al. 2025; Zhou et al. 2023] prove soundness of rewrites, never completeness; circuit optimisers [Kang and Oh 2023; Li et al. 2024; Paradis et al. 2024] search over rewrites; mathlib [The mathlib Community 2020, 2026], the AFP [Breitner 2010; Kharim et al. 2023] and MathComp [Gonthier et al. 2013] have no verified coset enumeration against a given concrete group.

Next are a verified matrix semantics closing the chain to $U_{p^n}(\mathbb{Z}[1/p, \omega])$, the qubit Clifford group [Selinger 2015], the qutrit rules [Li et al. 2025] as the case $p = 3$, and submission to the Agda standard library.

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A The Rules, the Normal Boxes, and the Size of the Development

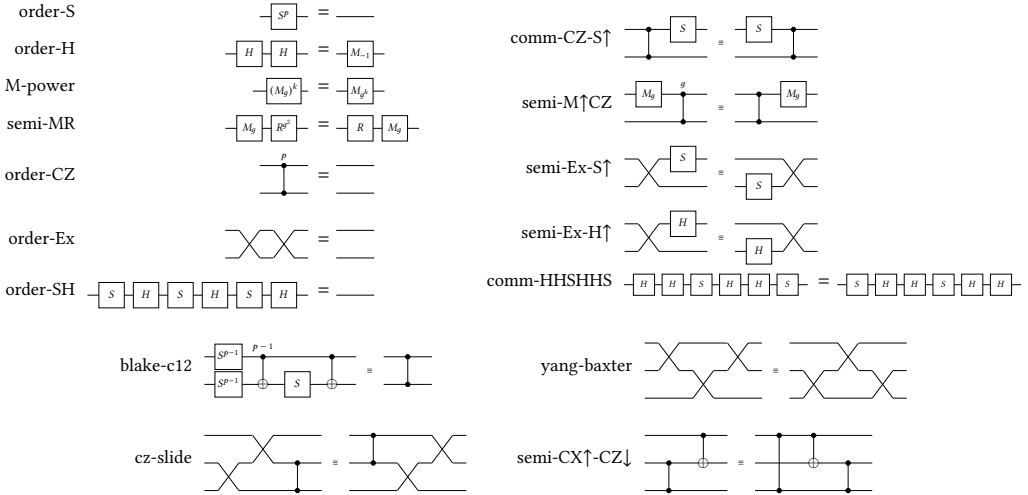


Fig. 1. The sixteen rules of `_CRel_` (§2) as circuit equations, drawn from the Agda terms. Diagrams read in matrix-multiplication order, so a word $w \bullet v$ is drawn with v on the left. `blake-c12` relates two circuits implementing controlled-Z; `yang-baxter` and `cz-slide` are the swap calculus. With the scalar restored (§5) only `order-SH` changes, to $(SH)^3 = \omega^{-1/8}$.

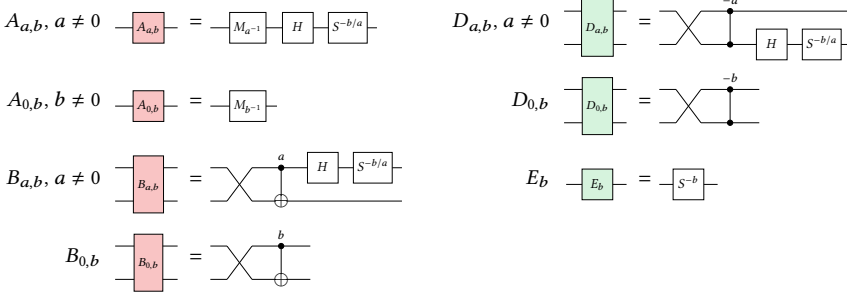


Fig. 2. The four kinds of normal box (the qubit paper’s Figure 6) and how each is spelled as a circuit by the section of the normal form (§4).

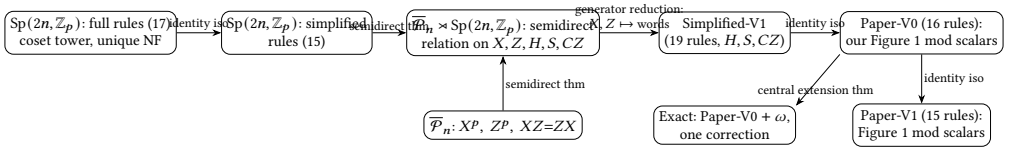


Fig. 3. The chain of presentations: every arrow is a verified presentation theorem or isomorphism of presentations; the semantic group is $\text{Sp}(2n, \mathbb{Z}_p)$ on the left, $\overline{\mathcal{P}}_n \rtimes \text{Sp}(2n, \mathbb{Z}_p)$ from the semidirect level on, and a central extension of it by $\langle \omega \rangle$ at “Exact”.

Table 1. Size of the development (lines of Agda incl. comments, wc -l, September 2026).

Component	Files	Lines
Framework (Word, Presentation, Circuit, Normalization, ForStdlib)	54	14 848
Symplectic group (normalisation 8 352; box relations 18 161; soundness etc. 8 886)	118	35 399
Projective Pauli group	3	1 220
Projective Clifford: semidirect, plumbing, shared calculus	26	17 996
Clifford with scalars	11	5 029
Other examples in the tree (symmetric, cyclic, wreath, Clifford+T, qubit Clifford)	107	27 552
Not on the chain (Simplified-V2), index, notations	10	4 344
Total	329	106 388

B Selected Agda Definitions

The normal-form datatype; the five hypotheses against which the library verifies a coset table (the general form of the soundness law `ract-sound` of Appendix C), in the generic `Transfer` module; the six one-qubit rules of the simplified symplectic rule set (§4), which with `order-CZ`, `comm-CZ-S`, `comm-CZ-S†` and `selinger-c10-c15` make its fifteen; and the conjugation action of the semidirect presentation (§5), the paper’s Lemmas 2.22–2.23 as words. All three are quoted from the development, as is the doubly inductive normal-form datatype of §4, shown first.

```

393
394  $M L ML : \mathbb{N} \rightarrow \text{Set}$ 
395  $M 0 = \top$ 
396  $M (1_+ n) = \text{Vec } D \ n \times E$ 
397  $L 0 = \top$ 
398  $L (1_+ n) = \text{Vec } B \ n \times A$ 
399  $ML 0 = \top$ 
400  $ML 1 = M 1 \times L 1$ 
401  $ML m @ (2_+ n) = M m \times L m$ 
402  $\quad \quad \quad \cup D \times ML (1_+ n)$ 
403
404  $(h =^{-1} f \text{-gen} : \forall (x : X) \rightarrow ([x]^w, l) \sim ((h^l) l (f x)))$  - 1
405  $(h \text{-wd-ax} : \forall (c : C) \{u t : \text{Word } Y\} \rightarrow u ==_2 t \rightarrow ((h^l) c u) \sim ((h^l) c t))$  - 2
406  $(f \text{-wd-ax} : \forall \{w v\} \rightarrow w ==_1 v \rightarrow (f^w) w \approx_2 (f^w) v)$  - 3
407  $([l] \approx \varepsilon : [1] \approx_2 \varepsilon)$  - 4
408  $(h = \text{ract} : \forall c b \rightarrow \text{let } (b', c') = h \ c \ b \text{ in } [c] \bullet [b]^w \approx_2 (f^w) b' \bullet [c'])$  - 5
409
410  $\text{order-S} : (1_+ n) \text{ SRel}, \quad S^{\wedge} p == \varepsilon$ 
411  $\text{order-H} : (1_+ n) \text{ SRel}, \quad H^{\wedge} 2 == M_{-1}$ 
412  $M\text{-power} : \forall k \rightarrow (1_+ n) \text{ SRel}, \quad Mg^{\wedge} k == M (g^{\wedge} k)$ 
413  $\text{semi-MS} : (1_+ n) \text{ SRel}, \quad Mg \bullet S == S^{\wedge} (g^* g) \bullet Mg$ 
414  $\text{semi-M}\uparrow\text{CZ} : (2_+ n) \text{ SRel}, \quad Mg \uparrow \bullet CZ == CZ^{\wedge} g \bullet Mg \uparrow$ 
415  $\text{semi-M}\downarrow\text{CZ} : (2_+ n) \text{ SRel}, \quad Mg \downarrow \bullet CZ == CZ^{\wedge} g \bullet Mg \downarrow$ 
416
417  $\text{conj} : \text{Sym.Gen } n \rightarrow \text{XZ.Gen } n \rightarrow \text{Word (XZ.Gen } n)$ 
418  $\text{conj Sym.H-gen } X\text{-gen} = Z$ 
419  $\text{conj Sym.H-gen } Z\text{-gen} = X^{\wedge} p-1$ 
420  $\text{conj Sym.S-gen } X\text{-gen} = X \bullet Z$ 
421  $\text{conj Sym.S-gen } Z\text{-gen} = Z$ 
422  $\text{conj Sym.CZ-gen } X\text{-gen} = X \bullet Z \uparrow$ 
423  $\text{conj Sym.CZ-gen } Z\text{-gen} = Z$ 
424  $\text{conj Sym.CZ-gen } (X\text{-gen } \uparrow) = X \uparrow \bullet Z$ 
425  $\text{conj Sym.CZ-gen } (Z\text{-gen } \uparrow) = Z \uparrow$ 
426  $\text{conj Sym.H-gen } (xz \uparrow) = [xz]^w \uparrow$ 
427  $\text{conj Sym.S-gen } (xz \uparrow) = [xz]^w \uparrow$ 
428  $\text{conj Sym.CZ-gen } (xz \uparrow \uparrow) = [xz]^w \uparrow \uparrow$ 
429  $\text{conj (sym } \uparrow) \quad X\text{-gen} = X$ 
430  $\text{conj (sym } \uparrow) \quad Z\text{-gen} = Z$ 
431  $\text{conj (sym } \uparrow) \quad (xz \uparrow) = \text{conj sym } xz \uparrow$ 

```

C An Introduction to the Framework

The formalisation is built on a library for presentations, normal forms and completeness of circuit relation sets [Anonymous 2026a,b]. This appendix introduces it on its smallest complete case study: circuits built from a single two-wire gate, the swap σ , which present the symmetric groups. The walkthrough exhibits the pipeline — gates, relations, cosets, tower, semantics, uniqueness, completeness, presentation — that §4 instantiates again for the symplectic factor, concretely enough that the latter can be read as the same pipeline with richer gates and a richer action; §5 glues such presentations together by the library’s product constructions.

C.1 Gates, generators, and circuits

A gate set is a family $\text{Gate} : \mathbb{N} \rightarrow \text{Set}$ giving, for each arity, the gates on that many adjacent wires; for permutations there is one gate, on two wires. The generic circuit layer of the library turns any such family into wire-indexed *generators*: a gate placed at the bottom of the wires, or a generator shifted up by one wire.

```

435  $\text{data Gate} : \mathbb{N} \rightarrow \text{Set where}$ 
436  $\sigma\text{-gate} : \text{Gate } 2$ 
437  $\text{data Gen} : \mathbb{N} \rightarrow \text{Set where}$ 
438  $\text{gate}_1 : \text{Gate } 1 \rightarrow \text{Gen } (1_+ n)$  — a 1-ary gate on the bottom wire
439  $\text{gate}_2 : \text{Gate } 2 \rightarrow \text{Gen } (2_+ n)$  — a 2-ary gate on the bottom two wires
440  $\uparrow : \text{Gen } n \rightarrow \text{Gen } (1_+ n)$  — the same generator, one wire higher
441

```


of the form $c \uparrow$. Read as permutations, two circuits lie in the same right coset of S_n exactly when they bring the same wire to the bottom position; there are $n+1$ choices, so S_n has index $n+1$, and a natural representative of each coset is the *descending staircase* of swaps that carries that wire down. In the library a coset is not a set of circuits but a label naming one, an element of the finite type $C\ n$; here the label is the length of the staircase, as a unary numeral bounded by the width, and the section $[_]^c$ maps it to the staircase circuit. The representatives on three, two, and one wires are drawn on the right:

data $C : \mathbb{N} \rightarrow \text{Set}$ where

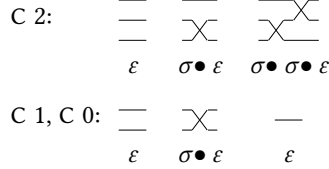
$\epsilon : C\ n$

$\sigma \bullet _ : C\ n \rightarrow C\ (1 + n)$

$[_]^c : C\ n \rightarrow \text{Circuit}\ (1 + n)$

$[\epsilon]^c = \epsilon$

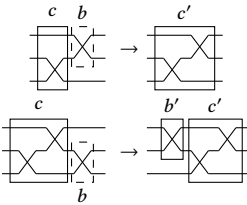
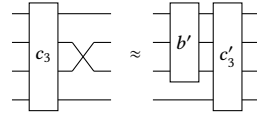
$[\sigma \bullet c]^c = \sigma \bullet ([c]^c \uparrow)$



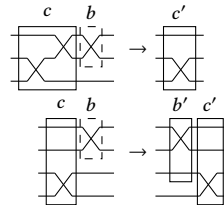
So $C\ n$ has exactly the $n+1$ elements drawn above for $n = 2, 1, 0$, matching the index of S_n in S_{n+1} . Why this yields a normal form is the classical argument of computational group theory [Holt et al. 2005; Sims 1994]: every $f \in S_{n+1}$ lies in one coset, say with representative c_n , so $f = f' \cdot c_n$ with $f' \in S_n$, and recursing on f' writes f uniquely as $c_1 c_2 \cdots c_n$ with $c_k \in C\ k$ — a numeral in the *factorial number system*, one digit per level, since $|S_{n+1}| = (n+1) \cdot n \cdots 2$.

Computing this factorisation is the job of the coset table. Given a coset, represented by its staircase, and one generator, the table says where their product lands — the updated coset — and what is left over on the wires above — the residual circuit — so that staircase followed by generator equals residual (shifted up) followed by updated staircase. Pushing σ into a staircase can lengthen it, cancel with it ($\sigma^2 = \epsilon$), or pass it through and shift it up (Yang–Baxter), or commute past; for a generator that does not touch the bottom wire we can recurse. The residual is empty or a single swap here, but in richer gate sets it can be a longer circuit, which is why the table returns a circuit on one wire fewer rather than a generator (Figure 4 draws the four cases):

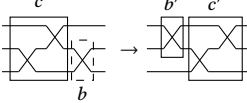
ract : $C\ (1 + n) \rightarrow \text{Gen}\ (2 + n) \rightarrow \text{Circuit}\ (1 + n) \times C\ (1 + n)$
 ract ϵ σ -gen = $\epsilon, \sigma \bullet \epsilon$ — grows
 ract $(\sigma \bullet \epsilon)$ σ -gen = ϵ, ϵ — cancels ($\sigma^2 = \epsilon$)
 ract $(\sigma \bullet \sigma \bullet c)$ σ -gen = $\sigma, \sigma \bullet \sigma \bullet c$ — passes (Yang–Baxter)
 ract ϵ $(g \uparrow)$ = $[g]^w, \epsilon$ — commutes past
 ract $(\sigma \bullet c)$ $(g \uparrow)$ = $\text{proj}_1(\text{ract } c\ g) \uparrow, \sigma \bullet (\text{proj}_2(\text{ract } c\ g))$ — recurses



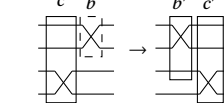
grows



cancels ($\sigma^2 = \epsilon$)



passes (Yang–Baxter)



commutes past

Fig. 4. Pushing a swap through a staircase: the four cases of the coset table ract as circuit rewrites.

σ -gen abbreviates gate₂ σ -gate. The last clause is the recursion: a shifted generator meets a staircase by meeting its tail one wire down. Each clause is certified by the *soundness law*, pictured on the right and proved once per table by a short equational derivation from the axioms:

ract-sound : $\forall c\ b \rightarrow \text{let } (b', c') = \text{ract } c\ b \text{ in } [c]^c \bullet [b]^w \approx b' \uparrow \bullet [c']^c$

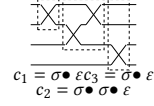
The stateful word traversal $_t$ extends ract to whole circuits, threading the coset from left to right and concatenating the residuals; run from the identity coset, it maps an $(n+2)$ -wire circuit to an $(n+1)$ -wire circuit and a coset — one level of normalisation.

Iterating the level is the coset tower, whose carrier grows by one digit per level:

$\text{NF} : \mathbb{N} \rightarrow \text{Set}$

$\text{NF } 0 = \top$

$\text{NF } (\text{suc } k) = \text{NF } k \times \mathbb{C} \, k - \top \times \mathbb{C} \, 0 \times \mathbb{C} \, 1 \times \cdots \times \mathbb{C} \, k$



so $\text{NF } n$ has exactly $n!$ elements. The normal-form function $\text{nf} : \text{Circuit } n \rightarrow \text{NF } n$ runs the tables; its section $\text{inv-nf} : \text{NF } n \rightarrow \text{Circuit } n$ rebuilds a circuit from its digits by concatenating the staircases, suitably shifted, as drawn on the right for $(\text{tt}, c_0, c_1, c_2, c_3) \in \text{NF } 4$, the digit $c_0 \in \mathbb{C} \, 0$ being the empty staircase — one staircase per level, each shifted up to the wires of its level: a “staircase of staircases”.

C.4 Semantics, uniqueness, and completeness

So far everything was syntax. The development interprets circuits in the group $\text{Permutation}'$ -group n of permutations of $\text{Fin } n$, with σ swapping the bottom two elements. *Soundness* — $w \approx v$ implies $[[w]] \doteq [[v]]$ — is a small case analysis over the axioms. The heart of the matter is the *unique normal form* property, that normal forms with equal denotations are equal: $[[\text{inv-nf } u]] \doteq [[\text{inv-nf } v]]$ implies $u \equiv v$. It is proved by induction on the tower: the lower digits fix the last point, so the top digit alone determines where the permutation sends it; equal denotations therefore force equal top digits, and cancelling them reduces the claim to the level below. Soundness and uniqueness are exactly the premises of the generic lemma by-normalization, which concludes *completeness*: $[[w]] \doteq [[v]]$ implies $w \approx v$. Together with surjectivity (every permutation is a product of adjacent transpositions), these assemble into the library’s *presentation record*: a Grouplike witness, which makes the presented monoid a group, and a proof that $[[_]]$ is a group isomorphism from that group onto the target ($\text{IsGroupIsomorphism}$ is the standard library’s, between the group the witness induces and G ; the record’s module plumbing that says so is elided):

```
record _IsPresentationOf_ (___ : WRel X) (G : Group a ℓ) : Set (a ⊔ ℓ) where
  field
    gl : Grouplike ___
    [[_]] : Word X → Group.Carrier G
    iso : IsGroupIsomorphism [[_]]
```

Every presentation theorem in the paper has the shape of this one:

$\text{symmetric-presentation} : \forall n \rightarrow (n \text{ VRel } _ == _) \text{ IsPresentationOf } (\text{Permutation}'\text{-group } n)$

read: the monoid of swap circuits modulo $\{\text{order}, \text{yang-baxter}\}$ and the structural rules is the group of permutations of $\text{Fin } n$. The symplectic factor of §4 is this pipeline again, with richer gates and a richer action, and §5 composes such presentations.